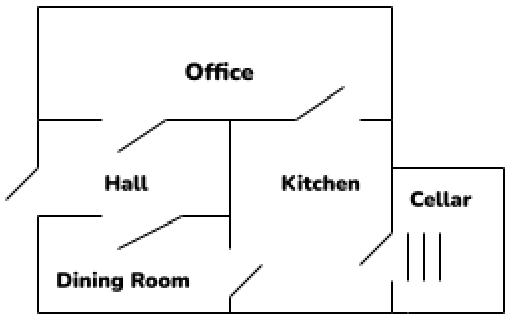
Assignment #7
CS3860

We are going to create a simple text-based adventure game. The premise of the game is that you are a 4-year old boy who is looking for their security blanket (blankie). The obstacle is a dark room. It's getting late and it is time for bed, so you need to find your blankie. Here is a map of the house.



Start by defining a predicate ***room/1*** that is true when its argument is one of the rooms given above. Now define a predicate ***location/2*** that tells us the location of things. Place the following items in the given rooms/locations :

- Place broccoli, crackers and an apple in the kitchen.
- place a desk and a computer in the office.
- place the washing machine in the cellar.
- place a flashlight in the desk
- place the blankie in the washing machine.

Define a predicate ***door/2*** that is true if a door exists between two rooms. This predicate has one shortcoming : door(office,hall) is a different fact from door(hall,office) . Address this by creating a predicate ***connect/2*** that is true if a connection between rooms exists. This will use the ***door/2*** predicate.

Write a predicate ***list_things/1*** that will list items in a place. You will need to use extra-logical predicates such as ***format/2*** , ***nl/0*** and ***tab/1***. Follow this with the definition of a predicate ***look_in/2*** that will tell you what items are located in a certain place.

Write a predicate ***list_connections/1*** that lists the rooms to which the argument (a room) is connected to. For example ***lists_connections(office)***. should return hall and kitchen.

Write a predicate ***can_go/1*** that will return true if you can go to a room from the room you are currently in. The predicate ***here/1*** tells you where you are (see below)

Finally, add the following predicates to complete the preliminary design :

1. a predicate ***edible/1*** that is true if something is considered edible (to a 4-year old).
2. a predicate ***tastes_yucky/1*** that is true if something is NOT edible
3. a couple of ground clauses, ***turned_off(flashlight)*** and ***here(kitchen)*** that initialize certain states.

PART 2.

We will continue to add predicates.

modify ***can_go/1*** so that it responds with an error message if you cannot move to a room from your current location (***here/1*** tells you that). Now write a predicate ***move/1*** that will update your location. It will remove the current ***here/1*** fact and replace it with an updated one.

Write a predicate ***look/0*** that will tell me what room I'm in, what I can see and what rooms I can move to. It will use predicates ***here/1***, ***list_things/1*** and ***list_connections/1***. Finally, add a predicate ***goto/1*** that will take you to a room (assuming you can go to that room) and have you look around. It will use predicates ***can_go/1*** , ***move/1*** and ***look/0***. At this point, you can test your game.

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-? goto(office).

you are in the office.
you can see:
      desk
      computer
you can go to:
      hall
      kitchen.

-?goto(cellar).

You can't get there from here.

-?goto(hall).

you are in the hall.
you can see:
you can go to:
      dining room
      office
```